

# Generation Interconnection Combined Feasibility / System Impact Study Report

## For Queue Project Q21-1762

Gaston Rd 100 kV  
November 02, 2022

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1762. The Interconnection Customer proposes a 268.2 MW solar photovoltaic facility interconnecting at the Gaston Rd 100 kV substation in the Indiana area. The request was received on May 26, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 530 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,343,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                       | Contingency                                        | Loading (%) | Rating (MVA) | DFAX (%) |
|------------------------------------------|----------------------------------------------------|-------------|--------------|----------|
| Teeland – Table Mt 100 kV                | Teeland – Table Mt 100 kV line                     | 103.5       | 252          | 47.2     |
| Teeland – Table Mt 100 kV Line           | Table Mt – Teeland 100 kV                          | 103.5       | 252          | 47.2     |
| Tortoise Substation – Teeland 100 kV #1  | Teeland – Tortoise 100 kV L                        | 103.4       | 223          | 45.2     |
| Teeland – Tortoise 100 kV                | Teeland – Tortoise 100 kV Cir. 1                   | 103.4       | 223          | 45.2     |
| Tortoise – Denver Terminal 100 kV Line   | Tortoise – Denver Terminal Substation 100 kV Ckt 2 | 103.2       | 201          | 44.5     |
| Tortoise – Denver Terminal 100 kV #2     | Denver Terminal – Tortoise 100 kV                  | 103.2       | 201          | 44.5     |
| N Mesquite – Denver Terminal 100 kV line | Parkrose – Twin Branch Pv 100 kV                   | 100.2       | 195          | 20.5     |

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

#### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

### 4. Required Network Upgrades and Cost Estimates

| Facility                                | Required Upgrade | Cost Estimate | Allocation to Project |
|-----------------------------------------|------------------|---------------|-----------------------|
| Teeland Substation – Table Mt 100 kV #1 | Reconductor line | \$250,000     | 31% (\$77,000)        |

| Facility                               | Required Upgrade           | Cost Estimate | Allocation to Project |
|----------------------------------------|----------------------------|---------------|-----------------------|
| Teelnd – Table Mt 100 kV Circuit 2     | Reconductor line           | \$269,000     | 31% (\$83,000)        |
| Teeland – Tortoise 100 kV Circuit 1    | Reconductor line           | \$711,000     | 32% (\$226,000)       |
| Teeland – Tortoise 100 kV              | Reconductor line           | \$1,036,000   | 32% (\$329,000)       |
| Tortoise – Denver Terminal 100 kV Line | Reconductor line           | \$647,000     | 32% (\$205,000)       |
| Tortoise – Denver Terminal 100 kV #2   | Reconductor line           | \$849,000     | 32% (\$269,000)       |
| Norh Mesquite – Denver Terminal 100 kV | Reconductor line           | \$661,000     | 23% (\$154,000)       |
|                                        | Total allocated to project |               | \$1,343,000           |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.